Assignment #9

1. What is the difference between a refrigerator and a heat pump? Explain each system by a diagram. Please include the related formula for each system if applicable.
2. In some North American locations such as Minnesota and Wisconsin, the heat pumps are not a good choice for heating dwellings? Explain. The answers from previous question can be a good source of information.

3 Air at 1600 K, 30 bar enters a turbine operating at steady state and expands adiabatically to the exit, where the temperature is 830 K. If the isentropic turbine efficiency is 90%, determine (a) the pressure at the exit, in bar, and (b) the work developed, in kJ per kg of air flowing. Assume ideal gas behavior for the air and ignore kinetic and potential energy effects.